

Lesson 6 Realize Data Reading and Writing based on EEPROM

6.1 Overview

In this lesson, we will introduce EEPROM and implement data reading and writing based on it.

6.2 Required Components

Components	Quantity	Picture
Adeept Robotic Arm Board	1	
Type-C USB Cable	1	
OLED Screen	1	

6.3 Principle Introduction

EEPROM (Electrically Erasable Programmable Read-Only Memory) is a type of non-volatile memory (power-loss persistent memory).

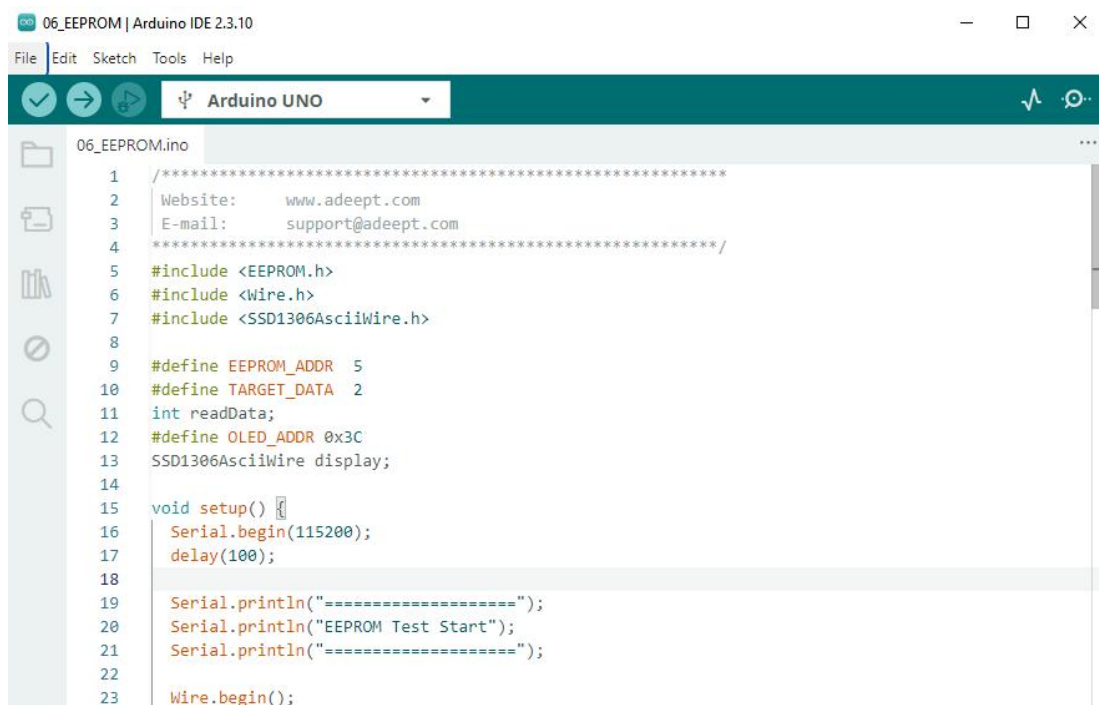
Unlike Arduino RAM, data stored in EEPROM will not be lost after power failure. Previously saved data can still be read even if the device is powered off or restarted.

Main features of EEPROM:

- Power-loss data retention: Data remains intact after power is cut off
- Rewritable: Stored contents can be modified and updated repeatedly
- Ideal for storing device parameters: such as calibration data, user settings, etc

6.4 Demonstration

1. Connect your computer and Adeept Robotic Arm Board with a USB cable.
2. Open "06_EEPROM" folder in "/Code" , double-click "06_EEPROM.ino" .



```
06_EEPROM | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
06_EEPROM.ino
1  /*****
2  Website:   www.adeept.com
3  E-mail:    support@adeept.com
4  *****/
5  #include <EEPROM.h>
6  #include <Wire.h>
7  #include <SSD1306AsciiWire.h>
8
9  #define EEPROM_ADDR  5
10 #define TARGET_DATA  2
11 int readData;
12 #define OLED_ADDR 0x3C
13 SSD1306AsciiWire display;
14
15 void setup() {
16   Serial.begin(115200);
17   delay(100);
18
19   Serial.println("=====");
20   Serial.println("EEPROM Test Start");
21   Serial.println("=====");
22
23   Wire.begin();
```

3. Select development board and serial port.

Board: **Tools---** > **Board---** > **Arduino AVR Boards---** > **Arduino Uno**

Port: **Tools** ---> **Port**---> **COMx**

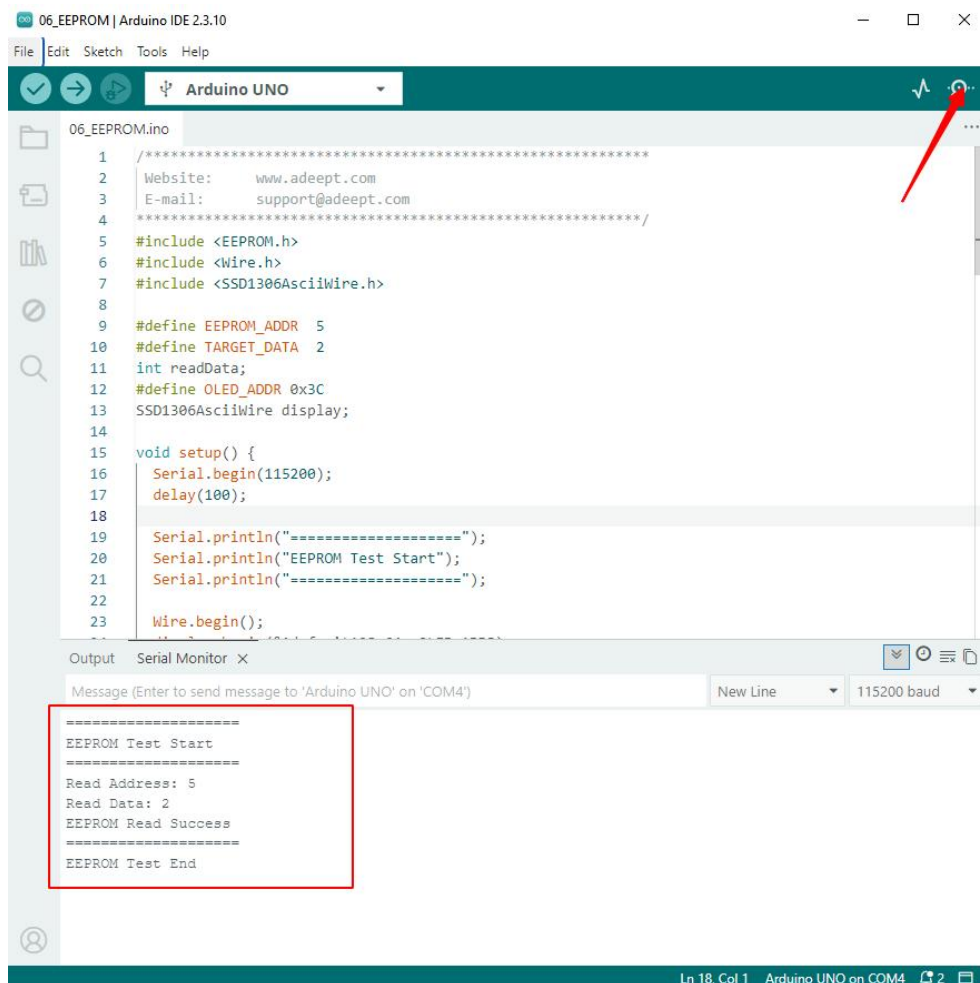
Note: The port number will be different in different computers.



4. After opening, click  to upload the code program to the Arduino. If there is no error warning in the console below, it means that the Upload is successful.

```
Output
Sketch uses 2952 bytes (9%) of program storage space. Maximum is 32256 bytes.
Global variables use 82 bytes (4%) of dynamic memory, leaving 1966 bytes for local variables. Maximum is 2048 bytes.
```

5. After the program runs successfully, you can open the Serial Monitor or view status information for EEPROM testing on the OLED screen.



6.6 Code

```
01  /*****
02  Website:      www.adeept.com
03  E-mail:       support@adeept.com
04  *****/
05  #include <EEPROM.h>
06  #include <Wire.h>
07  #include <SSD1306AsciiWire.h>
08
09  #define EEPROM_ADDR  5
10  #define TARGET_DATA  2
11  int readData;
12  #define OLED_ADDR 0x3C
13  SSD1306AsciiWire display;
14
15  void setup() {
16      Serial.begin(115200);
17      delay(100);
18
19      Serial.println("=====");
20      Serial.println("EEPROM Test Start");
21      Serial.println("=====");
22
23      Wire.begin();
24      display.begin(&Adafruit128x64, OLED_ADDR);
25      display.setFont(Adafruit5x7);
26      display.clear();
27
28      display.set2X();
29      display.setCursor(10,0);
30      display.print("EEPROM");
31      display.set1X();
32      display.setCursor(0,3);
33      display.print("Testing...");
34      delay(1000);
35
36      readData = EEPROM.read(EEPROM_ADDR);
37      Serial.print("Read Address: ");
38      Serial.println(EEPROM_ADDR);
39      Serial.print("Read Data: ");
40      Serial.println(readData);
41      if(readData == TARGET_DATA) {
42          Serial.println("EEPROM Read Success");
43          display.clear();
44          display.setCursor(0,0);
45          display.print("EEPROM OK");
46          display.setCursor(0,2);
47          display.print("ADDR:");
48          display.print(EEPROM_ADDR);
49          display.setCursor(0,3);
50          display.print("DATA:");
51          display.print(readData);
52      } else {
53          Serial.println("EEPROM Data Error");
```

```
54     Serial.println("Writing new data...");
55     display.clear();
56     display.setCursor(0,0);
57     display.print("EEPROM ERROR");
58     display.setCursor(0,2);
59     display.print("Writing...");
60     EEPROM.update(EEPROM_ADDR, TARGET_DATA);
61     delay(500);
62
63     readData = EEPROM.read(EEPROM_ADDR);
64     Serial.print("New Data:");
65     Serial.println(readData);
66     display.clear();
67     if(readData == TARGET_DATA) {
68         Serial.println("Write Success");
69         display.setCursor(0,0);
70         display.print("WRITE OK");
71     } else {
72         Serial.println("Write Failed");
73         display.setCursor(0,0);
74         display.print("WRITE FAIL");
75     }
76     display.setCursor(0,2);
77     display.print("ADDR:");
78     display.print(EEPROM_ADDR);
79     display.setCursor(0,3);
80     display.print("DATA:");
81     display.print(readData);
82 }
83 Serial.println("=====");
84 Serial.println("EEPROM Test End");
85 }
86
87 void loop() {
88
89 }
```